EduPulse AI — Model 3 Experiment B Report

Refined Recommendation Decision Rules | Generated: August 13, 2026

1. Dataset & Behavioral Methodology

Experiment B introduces explicit, priority-based behavioral rules to label learner profiles and eliminate class overlap while preserving realistic decision-boundary noise. The 100,000 synthetic records were evaluated using 8 priority-ordered rule conditions, achieving a balanced class distribution across all 8 recommendation actions.

2. V2 Model Evaluation Summary

Model	Accuracy	Precision	Recall	Specificity	F1 (W)	ROC AUC	MCC
Decision Tree	0.9546	0.9547	0.9546	0.9934	0.9546	0.9738	0.9479
Random Forest	0.9781	0.9788	0.9781	0.9969	0.9781	0.9971	0.9750
XGBoost	0.9766	0.9772	0.9766	0.9966	0.9766	0.9978	0.9732
LightGBM	0.9776	0.9781	0.9776	0.9968	0.9775	0.9977	0.9743
CatBoost	0.9768	0.9774	0.9768	0.9967	0.9768	0.9976	0.9735

3. Experiment A vs B Comparison

Metric	Experiment A (XGBoost)	Experiment B (Best)	Absolute Change
Accuracy	0.5256	0.9781	+0.4526
Weighted F1	0.5249	0.9781	+0.4532
Macro F1	0.5247	0.9789	+0.4542
ROC AUC	0.8982	0.9971	+0.0989
MCC	0.4572	0.9750	+0.5178
Specificity	0.9321	0.9969	+0.0648

4. Key Takeaways & Conclusion

The best V2 model is **Random Forest**, achieving a Weighted F1 score of **0.9781** and MCC of **0.9750**. Compared to Experiment A baseline (XGBoost), Accuracy improved by **+45.26%**, Weighted F1 by **+45.32%**, and MCC by **+0.5178**. Priority-based behavioral labeling successfully reduced class overlap while maintaining robust decision boundaries.